LAPRAS 5.6

COMPACT V HULL
Enhanced maneuverability
with a space-efficient design

6- CHANNEL RC SYSTEM
PID tuning for precise control
and stability

ADVANCED NAVIGATION
Accurate thrust vectoring and
responsive navigation

TEMPEST 2026 and Wavez 2026 (IITM)

Compact V-hull paired with a fixed propulsion setup ensures enhanced maneuverability.It'soperated through a 6-channel RC system using PID tuning, delivering accurate thrust vectoring and responsive navigation.

BUILDNEO

IN SEARCH OF INNOVATION

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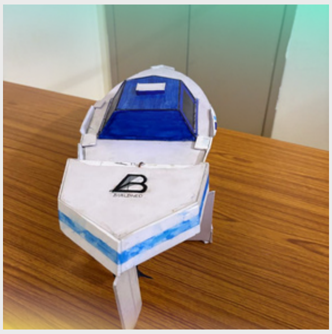
BUILDNEO: A student-led research and development initiative at the Indian Maritime University, Visakhapatnam, focused on innovation and practical skill development in naval architecture. Founded by ambitious alumni, it thrives on a network of past and present students collaborating on working prototypes and research. The initiative features a dynamic leadership team, including roles in external communications, electronics, technical documentation, coding, CAD modeling, fabrication, additive manufacturing, and social media. By promoting project flexibility and creativity, BuildNEO prepares members for academic and industrial challenges, aiming to become a hub of talent and innovation in the maritime industry.

LAPRACE 5.6

This project focuses on developing a compact, remotely controlled watercraft for short-range operations in controlled aquatic environments. It uses a high-efficiency brushless DC motor for reliable thrust with low power consumption. Wireless radio control allows precise navigation from a distance. The streamlined hull improves stability and movement on water. Internal components are arranged for efficient power use and smooth performance. The design also supports easy addition of sensors or payloads, making it suitable for environmental monitoring, aquatic research, and surface inspection in confined water areas.



OBJECTIVE :-

This study aims to conceive and build a small, high-performance, wireless-controlled vessel able to carry out limited-distance surface operations with steady performance under still-water environments. It showcases seamless coordination of thrust, energy supply, and guidance mechanisms in a minimal structure, and additionally establishes groundwork for enhancements like autonomy, information gathering, and sensing-driven route control for advanced future system capabilities expansion.

SPECIFICATIONS

- 43cm (Length) × 20.5 cm (Width) × 11 cm (Depth)
- Draft - 4cm
- 3-Blade Propeller
- A2212 1000KV BLDC motor,
- SimonK 30A ESC
- 11.1V 2200mAh 50C LiPo battery
- FlySky FS-iA6B 2.4GHz 6CH receiver
- Max speed - 5.6 knots
- Range: 600 m
- Endurance: 30–70 minutes

CONCEPT OF MODEL:-

This model is designed based on the concept of a lightweight high-speed boat. The idea is to reduce water resistance and achieve efficient forward motion using a propeller-driven system. The project demonstrates the practical application of marine engineering principles such as thrust generation, stability, buoyancy, and steering control.

SMART DESIGN HIGHLIGHTS:-

The bilge keels along both sides enhance stability and reduce rolling and wave resistance for smoother movement.A dedicated battery compartment ensures safety and organized power management.The balanced rudder improves steering control and maneuverability.A transverse bulkhead at the center supports the motor and minimizes vibration for better performance.The hull form was created using a minimum-cut method, reducing material waste and ensuring efficient construction.